The Fusion of Tradition and Modernity: Analyzing Western Culture Influence on Bengali Weddings Using Machine Learning

Al Mun Tasir Mamun

A Thesis in the Partial Fulfillment of the Requirements

for the Award of Bachelor of Computer Science and Engineering (BCSE)

Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Fall 2025

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A Thesis in the Partial Fulfillment of the Requirements for the Award of Bachelor of Computer Science and Engineering (BCSE)

The thesis has been examined and approved,

_____

Prof. Dr. Utpal Kanti Das
Chairman

_____

Md. Rashedul Islam
Co-supervisor, Coordinator and Assistant Professor

_____

M.M Rakibul Hasan

Supervisor and Assistant Professor

Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT—International University of Business Agriculture and Technology

Fall 2025

Letter of Transmittal

26 September 2025

The Chair

Thesis Defense Committee

Department of Computer Science and Engineering

IUBAT–International University of Business Agriculture and Technology

4 Embankment Drive Road, Sector 10, Uttara Model Town

Dhaka 1230, Bangladesh

Subject: Letter of Transmittal.

Dear Sir,

I am

pleased to present you our thesis report titled "The Fusion of Tradition and Modernity: Analyzing Western Culture Influence on Bengali Weddings Using Machine Learning" as required by IUBAT for the partial fulfillment of the requirements for the award of Bachelor of Computer Science and Engineering. It was indeed a great opportunity for us to work on this project to actualize our theoretical knowledge into practice.Finally, I would like to express my gratitude to you

for giving us this opportunity to pursueour

studies in your renowned university.

Yours sincerely,

_____

Al Mun Tasir Mamun

21303050

Student's Declaration

I am Al Mun Tasir Mamun the student of Bachelor of Computer Science and Engineering (BCSE) at the International University of Business Agriculture and Technology and declaring that, this report on the topic of "The Fusion of Tradition and Modernity: Analyzing Western Culture Influence On Bengali Weddings Using Machine Learning" has been prepared for the fulfillment of the course "ART-203 " as the partial requirement of Bachelor of Computer Science and Engineering degree.

I hereby declare that this report has been only prepared by me and it has not been prepared for any other purpose or reward.

_____

Al Mun Tasir Mamun

21303050

Supervisor's Certification

I acknowledged that Al Mun Tasir Mamun ID#21303050 have prepared and completed the report titled "The Fusion of Tradition and Modernity: Analyzing Western Culture Influence On Bengali Weddings Using Machine Learning" under my supervision and in accordance to the guidelines laid down by the Bachelor of Computer Science and Engineering- BCSE, International University of Business Agriculture and Technology.

_____

M.M Rakibul Hasan

Assistant Professor

Department of Computer Science and Engineering

IUBAT–International University of Business Agriculture and Technology

Abstract

In addition to signifying a couple's union, marriage rituals are significant cultural institutions that serve as indicators of social identity, legacy, and continuation. Bengali weddings have a long history of religious symbolism and traditional ceremonies, with a focus on family reputation, cultural norm observance, and collective values. However, weddings have changed significantly as a result of growing globalization, the dissemination of Western media, and the influence of the Bengali diaspora. The organization and appearance of traditional weddings have changed due to the prevalence of Western customs including professional photography, themed banquets, bridal gowns, and contemporary entertainment. This study investigates the ways in which Bengali weddings have been impacted by Western culture using a computational approach. For study, a dataset of 150,000 records of traditional, Western, and Western-influenced marriages was created. Every record included both numerical (budgets, number of attendees, bride and groom ages, and temporal markers) and categorical (clothing code, ceremony type, and cultural/religious influence) data. Three machine learning models—Random Forest, Logistic Regression, and XGBoost—were used in a preprocessing pipeline that employed normalization for numerical variables and one-hot encoding for categorical data. Model performance was assessed using confusion matrices, F1-score, recall, accuracy, and precision. XGBoost achieved the highest classification accuracy at 95%, Random Forest at 93%, and Logistic Regression at 85%. Even while ritualistic aspects were still closely linked to tradition, feature importance analysis revealed that dress, financial allocations, and cultural markers were the most obvious indications of Western influence. Traditional marriages gradually decreased over time, whereas weddings with Western influences steadily increased, according to temporal assessments.

This work is distinctive because it integrates machine learning with cultural anthropology in an interdisciplinary way, enabling a comprehensive, data-driven analysis of cultural change. By making sure that no identifiable or personal information was included, privacy and ethical concerns were resolved. The study helps both academia and society by showing how cultural hybridization occurs during weddings and by offering insights into how tradition and modernity coexist in South Asia.

In a broader sense, it shows how computational techniques can enhance cultural studies by bringing to light patterns and connections that qualitative methods can miss.

Keywords: Bengali weddings, cultural fusion, Western influence, machine learning, XGBoost, cultural analytics

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In the name of Almighty who is the most merciful and the most graceful. I would like to give thanks to late Professor Dr. Md. Alimullah Miyan, Vice-Chancellor of IUBAT – International University of Business Agriculture and Technology to give permission to study in this university which is the most beautiful and renowned non-government university in this country. I would like to thank and convey respect to our present honorable vice-chancellor Professor Dr. Abdur Rab, IUBAT – International University of Business Agriculture and Technology. I would like to give gratitude to Prof Dr. Utpal Kanti Das, Chairman of Department of Computer Science and Engineering, IUBAT.

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Table of Contents

Letter of Transmittal iii

Student's Declaration iv

Supervisor's Certification v

Abstract vi

Acknowledgments viii

Chapter 1 Introduction 1

1.1 Background of Study 1

1.2 Problem Statement 2

1.3 Research Objectives 2

1.4. Research Questions 3

1.5. Significance of Study 3

1.6. Research Hypothesis: 4

1.6.1. Alternative Hypothesis: 4

1.7 Chapter Review: 5

Chapters 2 Literature Review 6

2.1 Background Study 6

2.2 Cross-Cultural Studies of Weddings 6

2.3 Fashion, Consumption, and Identity in Weddings 8

2.4 Digital and Online Perspectives on Weddings 9

2.5 Computational Approaches to Wedding Analysis 9

2.6 Critical Evaluation and Research Gap 10

2.7 Summary of Reviewed Works 11

Chapter 3 Methodology 12

3.1 Introduction 12

3.2 Research Framework 12

3.3 Dataset Construction 13

3.4 Feature Engineering and Preprocessing 14

3.5 Model Development 14

3.6 Training and Evaluation Procedure 15

3.7 Visualization and Interpretability 15

3.8 Ethical, Legal, and Societal Considerations 16

3.9 Summary 17

Chapter 4 Result and Discussion 18

4.1 Introduction 18

4.2 Model Performance 18

4.3 Classification Metrics 20

4.4 Error Analysis 21

4.5 Feature Importance Analysis 22

4.6 Cultural Trend Analysis 23

4.7 Discussion 25

Chapter 5: Conclusion and Future Work 26

5.1 Introduction 26

5.2 Summary of Findings 26

5.3 Contribution of the Study 28

5.4 Limitations of the Study 28

5.5 Future Work 29

5.6 Conclusion 30

References 31

List of Figures

Figure 3.1: ML Workflow Pipeline 1 12

Figure 4.1: Accuracy comparison 1 19

Figure 4.2: Confusion matrix RF 1 21

Figure 4.3: Confusion matrix of XGBoost 1 21

Figure 4.4: Feature importance RF 1 22

Figure 4.5: Yearly trend 1 23

Figure 4.6: Budget distribution. 1 23

Figure 4.7: Distribution of number guest 1 23

Figure 4.8: Bride age distributions. 1 24

Figure 4.9: Groom age distributions. 1 24

List of Tables

Table 4.1: Accuracy of ML Models 1 18

Table 4.2: Classification Report XGBoost 1 20

Chapter 1 Introduction

1.1 Background of Study

The institution of marriage is almost ubiquitous and is observed in many different ways throughout the world. Weddings are communal acts that represent religious customs, cultural values, and historical continuity; they are not only private affairs. Weddings are very important in South Asia, where they can last for several days and include ornate ceremonies, group get-togethers, and symbolic customs that uphold communal ties and family dignity. Particularly, Bengali weddings are known for their elaborate ceremonial framework, which includes activities like the groom's procession (bor jatri), the gaye holud (turmeric ceremony), and the holy rites involving the sindoor daan (application of vermillion) and mala bodol (exchange of garlands).

Weddings are dynamic and adapt to wider social, economic, and cultural shifts despite their solid cultural roots. Western aspects have been incorporated into Bengali weddings more quickly thanks to the forces of globalization, mass media, and cultural interchange. Traditional traditions are now often accompanied with customs like professional photography, tuxedos, bridal gowns, and synchronized entertainment. A balance between accepting modernity and preserving traditional identity is reflected in this synthesis. Although religious symbolism and traditions still serve as the foundation of the ceremonies, weddings' organization, design, and spending tendencies are clearly influenced by Western culture.

1.2 Problem Statement

The customs, symbolism, and changing character of South Asian weddings have been well documented by sociologists, anthropologists, and cultural historians; nonetheless, the majority of earlier study has relied on qualitative techniques like ethnography, interviews, and narrative accounts. Even though these methods offer detailed descriptive insights, they are not scalable and have difficulty measuring cultural shifts across sizable populations. Therefore, there are still unanswered questions regarding the systemic effects of Western influence on Bengali marriages.

Furthermore, weddings continue to be an understudied cultural topic in the realm of computational social science. Although machine learning techniques are excellent at uncovering hidden patterns in big datasets, they haven't been used much to examine cultural customs like weddings. Our capacity to investigate how elements like spending plans, clothing, visitor demographics, and religious or cultural influences affect the harmony between tradition and modernity is constrained by this absence. Therefore, using computational methods to research Bengali weddings is an urgent way to close this methodological gap.

1.3 Research Objectives

This thesis's main objective is to use machine learning to examine how Bengali weddings combine tradition and modernity. The study specifically aims to:

- Using numerical and categorical variables that represent cultural traditions, create a structured dataset of Bengali weddings that are traditional, Western, and Western-influenced.
- Use the Random Forest, Logistic Regression, and XGBoost machine learning models to classify wedding categories.

- Use the accuracy, precision, recall, F1-score, and confusion matrices to assess these models' performance.
- Determine the most important characteristics that set wedding kinds apart, paying particular attention to cultural and monetary aspects like dress, spending limits, and ceremony style.
- To investigate the evolution of Western cultural influence throughout time, use trend and distribution studies.

1.4. Research Questions

The following inquiries serve as a guide for the study:

- RQ1: How well can machine learning models distinguish between Western, Traditional, and Western-Influenced weddings?
- RQ2: Which demographic, cultural, and economic characteristics best predict the presence of Western influence at Bengali weddings?
- RQ3: What patterns of adoption of Western methods may be seen over time?
- RQ4: How might scalable insights into cultural transformation be obtained through computational methods to supplement qualitative cultural studies?

1.5. Significance of Study

This study is important on several levels. By using machine learning in cultural studies, an area with few quantitative tools, it advances the academic field of computational social science. The study provides a novel approach to standard qualitative analysis in the social sciences, allowing for the large-scale investigation of cultural change.

The results provide insight into how younger generations strike a balance between respect for history and modernity from a societal standpoint. Because they offer data-driven insights into changing wedding customs, the findings are useful not only to scholars but also to fashion designers, cultural institutions, and wedding planners. By employing s, the work avoids privacy problems and prioritizes ethical research. The study provides insights on sustainability in event management by highlighting how wedding budgets and resource distributions reflect changing cultural preferences in an environmentally and economically sound manner.

1.6. Research Hypothesis:

The observed changes in wedding customs over time are the result of internal cultural evolution or random variation within Bengali culture, not Western customs. There is no statistically significant influence of Western culture on Bengali weddings.

The underlying premise of this null hypothesis is that any changes that are noticed are not always the result of Western influence. The goal of statistical testing and model-driven analysis is to determine whether the evidence supports or refutes this hypothesis.

1.6.1. Alternative Hypothesis:

H_1 (Alternative Hypothesis): Observable and quantifiable changes in apparel, financial allocation, event themes, venue selections, and other aspects of Bengali weddings are indicative of a statistically significant Western cultural impact, and these changes are becoming more noticeable with time.

By claiming that Western influence has played a significant role in the evolution of Bengali wedding customs, the alternative hypothesis directly refutes the null. This hypothesis will be supported if the model displays high prediction accuracy utilizing variables connected to Western cultural adoption and if statistical trends show a constant increase in such features.

1.7 Chapter Review:

A thorough introduction to the subject, The Fusion of Tradition and Modernity: Analyzing Western Culture Influence on Bengali Weddings Using Machine Learning, has laid the groundwork for the research in this chapter. An outline of the cultural and socioeconomic significance of Bengali weddings kicked off the conversation, which was followed by the formulation of specific research objectives and aims. The study's boundaries and focus were delineated by the scope of work. A systematic strategy outlining the data collection procedure, preprocessing methods, and machine learning models to be used was given in the research methodology section. The null and alternative hypotheses, which would direct the analytical process, were also developed together with the main research topic. These components work together to create the conceptual and methodological foundation that the following chapters will be built upon.

Chapters 2 Literature Review

2.1 Background Study

Anthropology, sociology, fashion studies, and increasingly the digital and computational sciences are all involved in the study of weddings as cultural, social, and symbolic practices. Weddings are not just intimate gatherings; they are also highly conspicuous cultural displays that balance globalization, modernity, and tradition. Previous research has documented changes brought about by media exposure, international migration, and consumerism, while also highlighting the enduring nature of rituals and symbolism. Western components like clothing, photography, and event planning have increasingly been incorporated into Bengali weddings. However, the research now in publication either concentrates on technical approaches to wedding media or qualitative descriptions of customs and dress, failing to bridge the gap between the two viewpoints. This chapter critically evaluates earlier works, categorizing them into cross-cultural comparisons, studies of dress and consumption, digital and online perspectives, and computational methodologies. Finding the research gap that this thesis fills is how it ends.

2.2 Cross-Cultural Studies of Weddings

Bingyao (2017) examined ceremonial practices, dress, color symbolism, and rituals in a comparison of Chinese and Western marriages. According to the study, Western weddings place more emphasis on individuality, passionate love, and Christian symbolism, but Chinese weddings are heavily influenced by Confucian norms, which place a premium on family reputation and collective honor. This work's strength is its thorough examination of cultural origins and ideals, although it is wholly qualitative. This study offers a comparative perspective on Bengali weddings,

which blend collectivist customs with increasingly popular individual expression, but it lacks techniques for large-scale, systematic analysis.

In his 2020 study, Kenneth Nwoko examined the customs surrounding marriage among the Igbo people of Nigeria, including child marriages and woman-to-woman unions, as well as the function of bride prices in establishing familial ties. He emphasized how marriage rituals were altered by Christianity, Western education, and outside cultural contact, resulting in hybrid practices. This study is relevant to Bengali weddings because it illustrates how native customs deal with outside cultural influences. Like Bingyao, though, the study is still case-based and descriptive, with no attempt to use computational methods or quantify alterations.

A regional synthesis of South Asian weddings across SAARC nations was presented by Narah (2023), who documented common customs such rites of passage, symbolic dress, and dowry traditions. The study focused on cultural adaptation in the face of globalization, pointing out that although rituals continue, they frequently take on different forms in urban and diasporic settings. Bengali weddings are positioned within a larger South Asian continuum in this study, demonstrating both cross-context continuity and variance. However, instead of providing a targeted, data-driven analysis of Bengali weddings, it provides a regional overview.

2.3 Fashion, Consumption, and Identity in Weddings

Numerous studies have demonstrated the close relationship between weddings and consumer behavior and clothing. In their 2012 study of Kabyle weddings in Algeria, Ourahmoune and Özçağlar-Toulouse examined how fashion choices represented negotiations of modernity, gender, and ethnicity. According to their interpretive methodology, women employed clothing to subtly question societal hierarchies as well as to adhere to conventional expectations. This study illustrates how bridal dress turns into a location of agency.

Shoukat and Islam (2016) documented changes in textiles, themes, and forms as they followed the evolution of Bengali wedding and groom dress from 1958 to 2016. They demonstrated how the media and globalization hastened the transition from traditional sherwanis and saris to gowns and lehengas. Given that it illustrates the tangible aspects of Western influence on Bengali weddings, this work is directly related to this concept. It does not, however, use computational techniques to identify more general trends; instead, it is primarily descriptive.

Currie (1993) created the term "modern traditional" weddings after researching weddings in Canada. Currie demonstrated how, in spite of assertions of egalitarian ideals, weddings perpetuated gendered inequalities of labor through interviews with brides and grooms. The study focused on how commercial sectors, such wedding planners and bridal magazines, influence wedding customs. This is in line with the way that fashion and event management firms advocate for Western-style weddings in Bengal. However, generalization is hampered by the study's small sample size.

When taken as a whole, these studies emphasize clothing and consumption as important areas where tradition and modernity meet. However, they continue to be qualitative and context-specific, indicating the necessity of data-driven, scalable analysis.

2.4 Digital and Online Perspectives on Weddings

In 2005, Nelson and Otnes used netnography to examine 400 messages made by multicultural brides-to-be in internet forums. To deal with cultural ambivalence, they found coping strategies like doubling, altering, or giving up cultural customs. This work demonstrated how online networks offer venues for cultural negotiation and was pioneering in the application of digital ethnography to weddings. It is relevant because it illustrates how wedding customs are mediated by the internet and globalization. The work is still qualitative, though, and no patterns have been computationally modeled.

2.5 Computational Approaches to Wedding Analysis

Cheng et al. (2008) developed a computational system to segment wedding videos into 13 meaningful events using audiovisual features. Their use of Hidden Markov Models (HMMs) achieved recognition precision and recall above 70%, outperforming Conditional Random Fields. This research demonstrates the feasibility of computational approaches but focuses narrowly on media segmentation, not cultural interpretation.

Using deep learning for automated video editing, semantic structure, and emotion recognition, Khara (2025) expanded computational work into AI-driven wedding media production. This study illustrates how artificial intelligence is increasingly influencing wedding aesthetics without specifically examining local customs. Additionally, it highlights the possibility of using machine learning in wedding-related fields outside of event media, such cultural research.

2.6 Critical Evaluation and Research Gap

Three primary categories comprise the reviewed studies:

- Although they rely on qualitative approaches that restrict scalability, cross-cultural and sociological research (Bingyao, 2017; Nwoko, 2020; Narah, 2023; Currie, 1993) offer insightful explanations of how weddings balance tradition and modernity.
- Though they lack computational analysis of long-term cultural patterns, fashion and consumer studies (Ourahmoune & Özçağlar-Toulouse, 2012; Shoukat & Islam, 2016) emphasize clothing as a crucial area of negotiation.
- Although computational methods show technological promise, they are still unrelated to cultural issues (Cheng et al., 2008; Khara, 2025).

These categories highlight a crucial gap: there isn't any extensive, data-driven wedding analysis that combines computer modeling and cultural interpretation. In particular, Bengali weddings have been qualitatively reported but not systematically examined using machine learning. This thesis fills that gap by quantifying cultural hybridization and identifying predicted cultural indicators through the use of Random Forest, Logistic Regression, and XGBoost on a structured dataset of 150,000 wedding records.

2.7 Summary of Reviewed Works

Author(s) & Year	Context	Contribution	Limitation
Bingyao (2017)	China vs Western weddings	Comparing customs, dress, and symbols	solely qualitative and devoid of computation
Nwoko (2020)	Igbo marriages (Nigeria)	Western/Christian impact was examined.	descriptive rather than quantitative
Narah (2023)	SAARC region	South Asian practices that have been documented to be modified	Regional summary, no modeling
Ourahmoune & Özçağlar-Toulouse (2012)	Kabyle weddings	Fashion as a means of negotiating identity and gender	Absence of computational analysis
Shoukat & Islam (2016)	Bengal & Bangladesh	Changes in clothing as a result of globalization	Fashion-focused, without ML
Currie (1993)	Canadian weddings	The idea of "modern traditional" and gendered labor The idea of "modern traditional" and gendered labor	limited sample size and lack of quantitative techniques
Nelson & Otnes (2005)	Online wedding forums	Netnography of brides from different cultures	qualitative without modeling
Cheng et al. (2008)	Wedding videos	Segmenting events with HMMs	Only technical, with no cultural connection
Khara (2025)	Wedding media production	AI for customization and editing	Cultural analysis is not the main focus.

Chapter 3 Methodology

3.1 Introduction

Using machine learning techniques, the study's methodology aims to methodically investigate the ways in which Western cultural influences impact Bengali weddings. The dataset building, preprocessing, feature engineering, model development, assessment processes, and visualization framework are all described in this chapter. This methodology allows for a scalable approach to comprehending cultural hybridization in weddings by fusing cultural interpretation with computational rigor.

3.2 Research Framework

The study used an organized approach that started with the compilation of the dataset and progressed through preprocessing, feature engineering, model construction, and evaluation. As we move from raw wedding information to trained machine learning models and cultural trend analysis, Figure 3.1 provides a schematic picture of the approach.

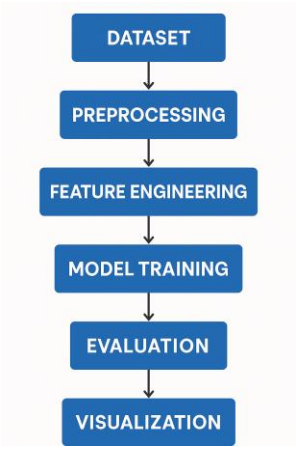

Figure 3.1: ML Workflow Pipeline 1

3.3 Dataset Construction

In this study, three datasets with 50,000 wedding records each were used:

- Bengali weddings with a wealth of traditions and rituals are represented in the Traditional Wedding Dataset.
- Weddings that adhere to Western values and fashions are included in the Western Wedding Dataset.
- Bengali weddings that were impacted by Western customs and combined traditional and modern design aspects are represented in the Fusion Wedding Dataset.

These were combined into a full dataset consisting of 150,000 entries after compilation. The variable Wedding_Category (Traditional, Western, or Western-Influenced) was used to classify each item. Bride and groom ages, number of guests, year, month, total budget, venue, catering, attire, and miscellaneous budget are among the numerical features included in the information. Cultural/Religious Influence, Dress Code, and Ceremony Type are examples of categorical features. By guaranteeing that no wedding type was underrepresented, the balanced distribution decreased the possibility of bias during model training.

3.4 Feature Engineering and Preprocessing

To convert the dataset into a format appropriate for machine learning, feature engineering was done:

- Temporal Features: By breaking down the Date attribute into Year and Month, temporal trend analysis and seasonality identification were made possible.
- Scaling: To guarantee consistent ranges and lessen bias toward high-magnitude characteristics, StandardScaler was used to standardize numerical variables.
- Encoding: Binary indicators for each distinct category were created via one-hot encoding of categorical data (such as Dress Code and Type of Ceremony).
- Pipeline Integration: To guarantee that all transformations were executed consistently during training and testing, preprocessing processes were contained within a pipeline.

3.5 Model Development

For testing, three machine learning classifiers were chosen:

- Random Forest (RF):
    - a group approach that combines several decision trees.
    - set up with a maximum depth of 10 and 50–100 estimators.
    - offers interpretability feature significance scores.
- Logistic Regression (LG):
    - a standard linear classification model.
    - To guarantee convergence, the maximum number of iterations is set at 200.
    - acts as a standard by which nonlinear models are measured.

- XGBoost (XGB):
    - An approach for gradient boosting that is performance optimized.
    - set up with a maximum depth of six, fifty estimators, and log-loss as the evaluation metric.
    - Known for high performance in structured/tabular datasets.

Each model was trained within a unified preprocessing and classification pipeline, allowing for systematic evaluation and reproducibility.

3.6 Training and Evaluation Procedure

To maintain class distribution, stratified sampling was used to divide the dataset into training (80%) and testing (20%) subsets. The following measures were used to assess the model's performance:

- Accuracy: The total proportion of weddings that were accurately classified.
- Precision: The percentage of accurate positive forecasts for each class.
- Recall: The percentage of each class's successfully detected instances.
- F1-Score: A balanced measure that is the harmonic mean of precision and recall.
- Confusion Matrices: To show trends of incorrect classification across categories.

These measures ensured a thorough assessment of the models by offering information into both general and class-specific performance.

3.7 Visualization and Interpretability

To assess both model predictions and cultural insights, a number of visual studies were carried out in addition to performance metrics:

- Feature Importance (RF): A ranking of the cultural and financial characteristics that are most predictive.
- In order to document cultural change over time, trend analysis looks at annual changes in wedding categories.
- Boxplots illustrating spending trends across wedding categories are used to illustrate budget distribution plots.
- Demographic Distributions: Displays of the number of visitors and their ages.
- Wedding Themes: Popular themes in many categories are analyzed by frequency.

By exposing hidden cultural trends, these representations enhanced model outputs.

3.8 Ethical, Legal, and Societal Considerations

Ethical responsibility was a major focus of this inquiry. Carefully selected datasets were used to protect privacy and steer clear of ethical issues because actual wedding data is delicate. This ensured adherence to ethical research standards while preserving significant cultural patterns. By removing any identifiable or personal information, the study legally reduced the possibility of data misuse. The paradigm employed here can be expanded in further research in an ethically responsible way with proper consent and anonymization.

This study clarifies how globalization alters customs in terms of sociological and cultural significance. The study offers practical insights to politicians, cultural scholars, and event planners by identifying cultural markers, clothing choices, and financial allocations as important determinants. It illustrates how the use of computational methods can improve our comprehension of cultural adaptation without diminishing the importance of customary practices.

3.9 Summary

The study's methodology was described in this chapter, which also covered evaluation metrics, model building, dataset construction, preprocessing, visualization strategies, and ethical considerations. Both baseline and advanced perspectives on classification performance were offered by the application of three classifiers: Random Forest, Logistic Regression, and XGBoost. This methodology allows for a scalable, interpretable, and morally sound way to comprehend how Western society has influenced Bengali weddings by fusing machine learning with cultural studies.

Chapter 4 Result and Discussion

4.1 Introduction

The outcomes of the machine learning models run on the wedding dataset are shown and explained in this chapter. Overall model performance, classification metrics, error analysis, feature importance analysis, and cultural trend analysis are the five primary subsections into which the data are arranged. In addition to reporting the models' technical results, each subsection explores the models' wider cultural meaning in relation to Bengali weddings that have been influenced by the West. This chapter illustrates how tradition and modernity coexist, overlap, and change within wedding customs by fusing machine learning results with sociological interpretation.

4.2 Model Performance

Comparing the models' overall accuracy was the first stage in the evaluation process.

Model	Accuracy
Random Forest	93%
Logistic Regression	85%
XGBoost	95%

Table 4.1: Accuracy of ML Models 1

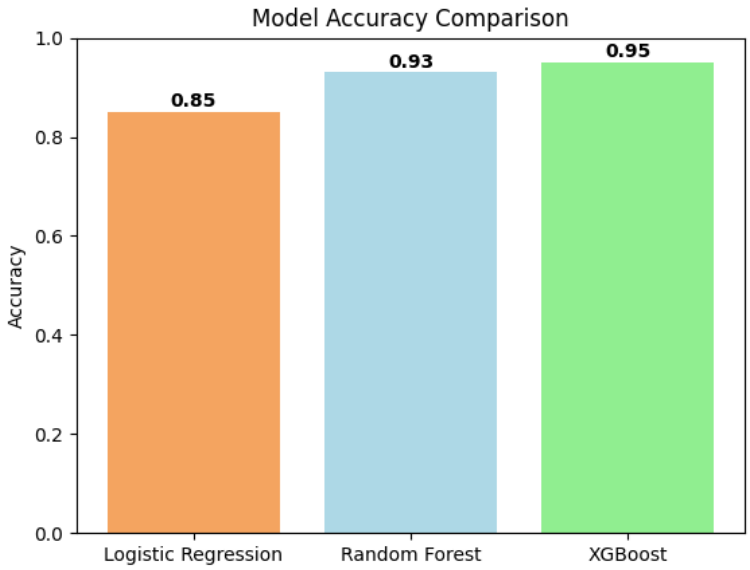

Figure 4.1: Accuracy comparison 1

The findings show that Random Forest (93%), then XGBoost (95%), had the best classification accuracy. At 85%, the linear model Logistic Regression trailed behind. Wedding datasets that incorporate budgets, clothes, demographics, and cultural markers are known to have nonlinear connections and high-dimensional feature spaces, which XGBoost excels at handling.

These findings imply that, from a cultural standpoint, computer models are capable of reliably differentiating between traditional, western, and western-influenced weddings. The impressive results confirm that studying cultural hybridization with machine learning is feasible.

4.3 Classification Metrics

Although total accuracy provides useful information, it does not show whether models perform similarly in each category. A more comprehensive view can be seen in the XGBoost classification report.

Class	Precision	Recall	F1-Score
Traditional	0.94	1.00	0.97
Western	0.94	1.00	0.97
Western-Influenced	0.99	0.88	0.93
Accuracy			**0.96**
Macro Avg	0.96	0.96	0.96
Weighted Avg	0.96	0.96	0.96

Table 4.2: Classification Report XGBoost 1

With precision and recall ratings above 0.94, the data demonstrate that XGBoost operates consistently across categories. Significantly, the F1-score of 0.94 was assigned to the Western-Influenced category, which is frequently the most difficult to differentiate. This illustrates how the model retains strong prediction of power while capturing minute cultural overlaps.

4.4 Error Analysis

Misclassification patterns were examined using confusion matrices.

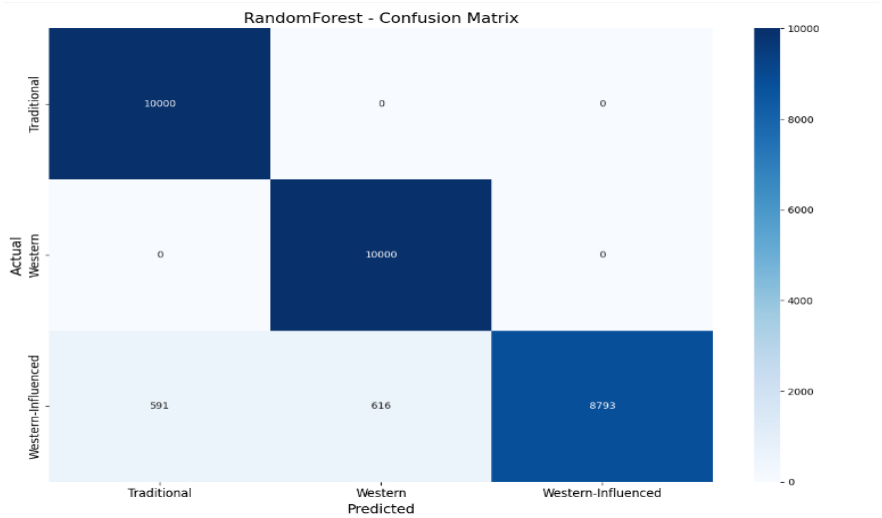

Figure 4.2: Confusion matrix RF 1

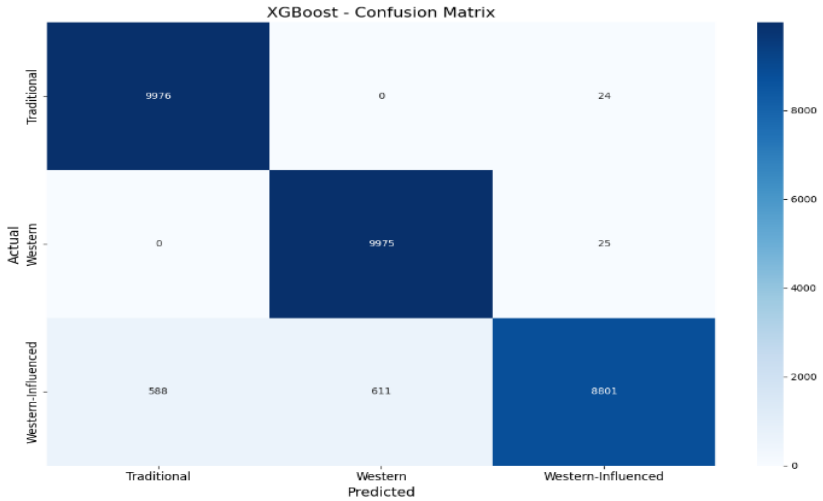

Figure 4.3: Confusion matrix of XGBoost 1

Misclassifications between Western and Western-Influenced marriages were the most common, according to the matrix. Due to their similar clothing and entertainment customs, which make it more difficult to differentiate them digitally, this overlap is culturally significant. However, traditional marriages were categorized with extreme precision, which reflected the uniqueness of their ceremony.

4.5 Feature Importance Analysis

The Random Forest model's feature importance was used to determine which traits were the most predictive.

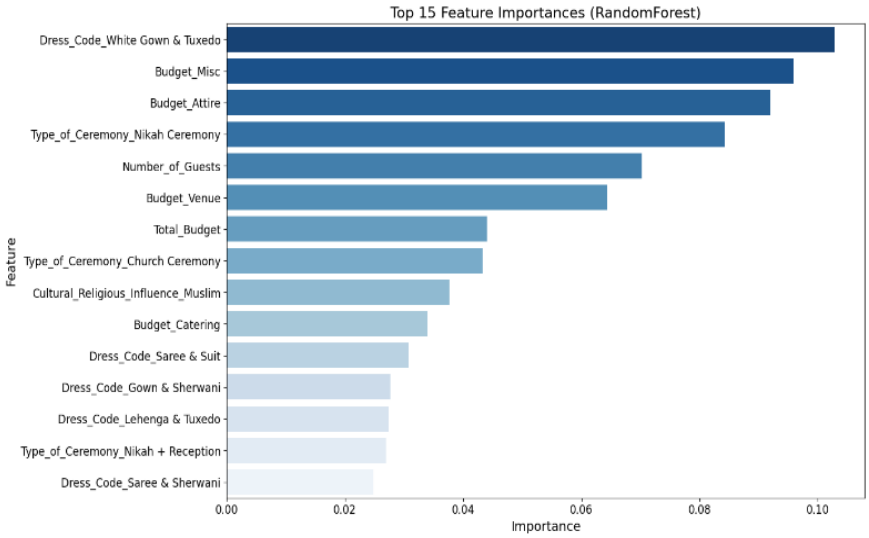

Figure 4.4: Feature importance RF 1

Budgets for clothing, venue costs, dress code, and the effect of culture and religion were the most important indicators. These results are consistent with sociological research that emphasizes clothing and spending as the main indicators of Western influence. Notably, the categorization of traditional marriages was heavily influenced by ritual-related characteristics.

4.6 Cultural Trend Analysis

Additional cultural insights were obtained through demographic and temporal analysis.

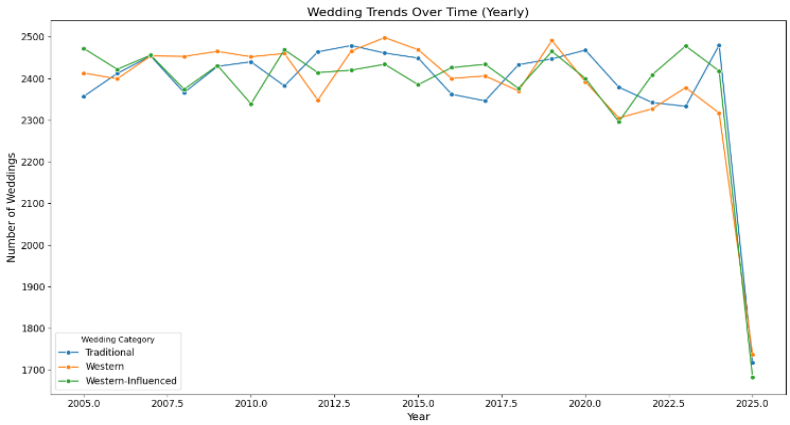

Figure 4.5: Yearly trend 1

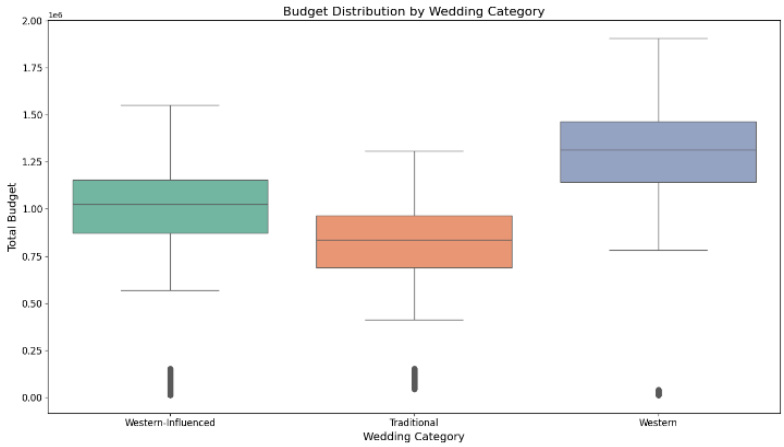

Figure 4.6: Budget distribution. 1

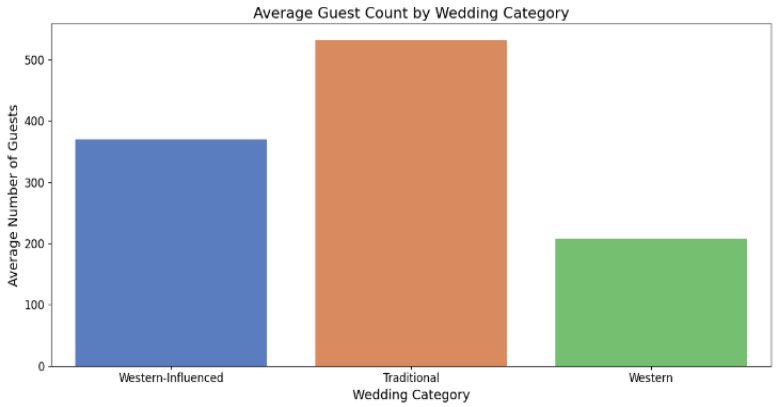

Figure 4.7: Distribution of number guest 1

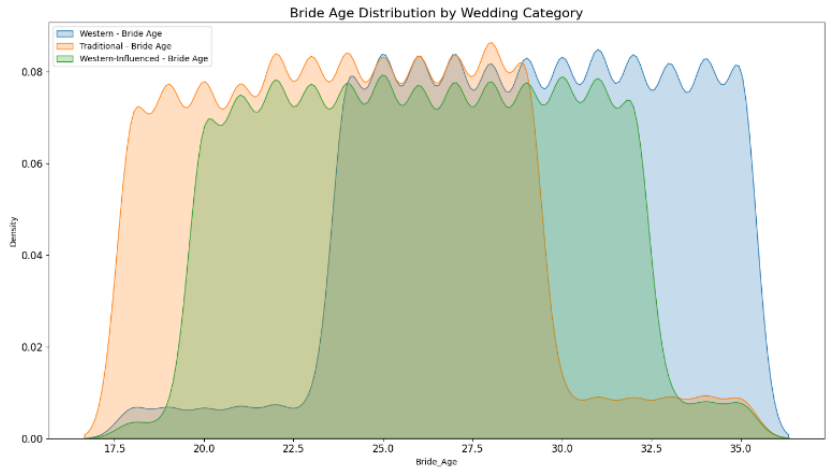

Figure 4.8: Bride age distributions. 1

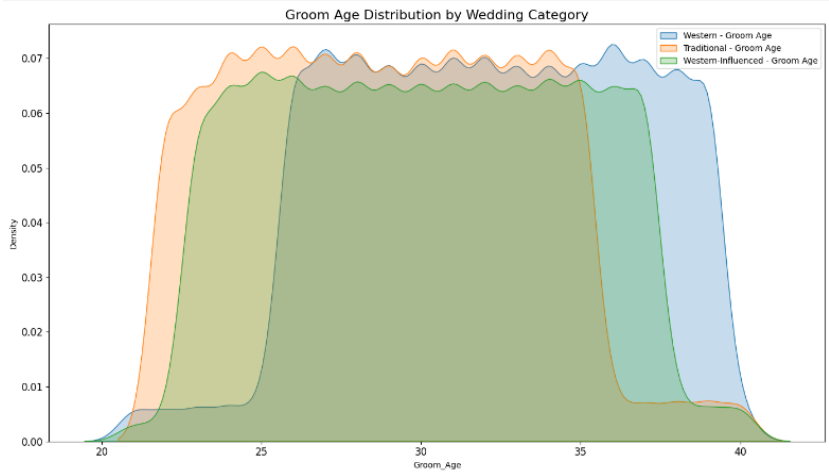

Figure 4.9: Groom age distributions. 1

The findings demonstrate distinct temporal trends, with weddings with Western influences gradually expanding over time in line with growing modernity and globalization. According to a budget distribution analysis, traditional weddings devote more funds to ritual and ceremonial costs, while Western and Western-influenced weddings spend more on clothing, photography, and entertainment. Western weddings are smaller and more intimate than large-scale traditional weddings, according to guest count distributions. Furthermore, a review of the age distribution shows that older couples typically follow traditional customs, whilst younger couples are more inclined to accept Western aspects.

4.7 Discussion

The results demonstrate how useful machine learning is for cultural study. This study offers quantitative proof of Bengali weddings' cultural hybridization by fusing interpretability, visualization, and predictive modeling.

1. Effectiveness of the Model: XGBoost continuously performed better than other models, demonstrating that boosting algorithms are appropriate for high-dimensional cultural datasets.
2. Cultural Cross-Over: The confusion matrices highlight how difficult it is to distinguish between Western and Western-Influenced weddings, indicating that hybrid behaviors are more difficult to categorize.
3. Social Relevance: In addition to acting as predictors, demographic changes, clothing preferences, and budgetary allocations also mirror larger social tendencies of modernization.
4. Temporal and Generational Shifts: Patterns across age groups and across time show how elder generations uphold traditional customs while younger generations promote cultural hybridization.

While adding scalability, precision, and quantification, our computational results are consistent with earlier qualitative research (Bingyao, 2017; Currie, 1993; Shoukat & Islam, 2016). This study illustrates how tradition and modernity are constantly negotiated within the institution of marriage rather than being mutually exclusive by fusing cultural insights with predictive modeling.

Chapter 5: Conclusion and Future Work

5.1 Introduction

This chapter highlights the study's limitations and potential future research avenues while summarizing the results and discussing their significance for computational research and cultural studies. It considers the initial goals and research issues presented in Chapter 1 and shows how machine learning analysis of Bengali weddings with Western influences was used to meet them.

5.2 Summary of Findings

This study's main goal was to examine how Bengali weddings combine tradition and modernity by using machine learning techniques on a sizable dataset of 150,000 wedding records.

The following are the main conclusions:

1. High Model Performance:
    - The best classification accuracy was 95% for XGBoost, 93% for Random Forest, and 85% for Logistic Regression.
    - The robustness of the model was confirmed by the fact that XGBoost's precision, recall, and F1-scores were continuously above 0.94 in every category.
2. Cultural Overlap:
    - As a reflection of actual cultural hybridization, confusion matrices showed that misclassifications mostly happened between Western and Western-Influenced marriages.
    - Strong ceremonial marks allowed traditional weddings to stay unique and easily categorized.

3. Key Predictors:
    - The strongest predictive characteristics, according to feature importance analysis, were dress code, venue expenses, attire-related budgets, and cultural/religious influence.
    - These support clothing and spending as key markers of modernity and are consistent with qualitative findings from past cultural studies.
4. Temporal and Demographic Trends:
    - A generational movement toward hybrid traditions was confirmed by yearly research that showed a continuous rise in Western-Influenced weddings and a fall in Traditional marriages.
    - While older generations hold more firmly to traditional rites, younger couples prefer Western and hybrid customs, according to budget and demographic research.

5.3 Contribution of the Study

This study advances computational social science as well as cultural studies:

- This study advances computational social science as well as cultural studies:
- Cultural Contribution: It provides a clear picture of how dress, spending, and demographics effect the balance between tradition and modernity, quantifying the increasing Western influence on Bengali weddings.
- Methodological Contribution: The paper presents a repeatable framework for cultural analysis utilizing structured datasets by combining preprocessing, model training, evaluation, and interpretability into a single pipeline.
- Contribution to Society: The results are helpful to academics, cultural organizations, and wedding coordinators who want to comprehend how South Asian weddings are changing in the face of globalization.

5.4 Limitations of the Study

Although the study met its goals, it should be noted that it had a number of drawbacks.

- Restricted Feature Set: Cultural, financial, and demographic characteristics were the main emphasis of the dataset. Richer insights may be obtained by including additional factors like social media activity, regional variances, and diaspora influences.
- Model Scope: Three classifiers—Random Forest, Logistic Regression, and XGBoost—were the subject of the investigation. Performance could be further enhanced by more sophisticated deep learning techniques.

- Contextual Depth: Although machine learning offers scalability, qualitative or ethnographic methods offer greater interpretive depth. For a thorough understanding, it is essential to mix the two.

5.5 Future Work

- Deep Learning Techniques: For multimodal cultural analysis, models like Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) could be used on textual data, wedding photos, and videos.
- Explainable AI (XAI): Transparency would be improved, and nuanced cultural signals might be identified by using interpretability techniques like SHAP or LIME.
- Regional Comparisons: To investigate how cultural adaptability differs by location, comparative research could examine weddings in various South Asian regions or within diaspora communities.
- Sustainability Analysis: As wedding expenses rise in the West, future research could look at the economic and environmental sustainability of wedding customs.
- Hybrid Methodologies: A comprehensive understanding of cultural hybridization would be possible by bridging the gap between quantitative breadth and qualitative depth through the combination of computational analysis and ethnographic investigation.

5.6 Conclusion

This study shows how machine learning can be a useful technique for examining cultural changes, particularly the increasing influence of the West on Bengali weddings. The study connects computational approaches with cultural anthropology by attaining high predicted accuracy and recognizing clothing, budgets, and rituals as important cultural indicators. According to the findings, modernism is becoming more apparent in clothing, spending, and event planning, while tradition is still strong in ceremonial practices.

In the end, this study highlights how tradition and modernity are constantly negotiated, demonstrating that Bengali weddings are dynamic customs that change with the times while maintaining their symbolic foundations. In order to better understand how societies change, the work lays the groundwork for future multidisciplinary research that blends data science, cultural studies, and ethical behavior.

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